a sleeve surrounding the jaws. This movement enables the vice to hold small round objects. Strings and wires of small diameters can be held with a pin vice.



UNIVERSAL VICE

The universal vice can be swivelled in a horizontal plane similar to a swivel vice and can also be tilted in any vertical position for angular cuts.



MAINTENANCE OF VICE

1. Vices should be maintained properly. Care should be taken that the screw of the vice is free from dirt or metal burrs.
2. The screw of the vice should be lubricated with grease for proper sliding of the movable jaws.
3. The top of the vice should not be used as an anvil.

2.4 FILE

File is a hardened steel tool having slanted and parallel rows of cutting edges or teeth on its surface. It is used to cut, smooth or fit metal parts. It is also used on wooden and plastic parts. It cuts all materials expect hardened steel. Small quantities of unrequired metal can be removed with files.



File

Metal burrs left out after chiselling and backsaws cutting are removed with the help of files. It is also used to sharpen the cutting edges of sharp tools like saws. The tang is a pointed part which fits into the handle. The point is the end opposite to the tang. The heel is next to the tang. The face of the file has a slanting row of cutting edges. File is made of high carbon steel.

Files are classified according to the following factors.

1. Effective length
2. Sectional form
3. Cut of teeth
4. Grade

Size of the file

The length of the file is its size. It is measured from the point to the

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